

NESTED FOR LOOP

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WHAT IS A LOOP?

A loop is used to repeat a block of code multiple times.
It helps reduce code repetition.

Types of loops in Python :

- for loop
- while loop
- Nested loop

WHAT IS A NESTED LOOP

- A loop inside another loop is called a nested loop
The inner loop runs completely for each iteration of the outer loop

SYNTAX

```
For i in range(n):  
    for j in range(m):  
        # code block  
  
print(i,j)
```

HOW IT WORKS

1. Outer loop runs first.
2. For each outer loop iteration:
 - Inner loop runs completely.
3. This continues until outer loop finishes.

REAL-LIFE EXAMPLE

Used for:

- Printing patterns stars, numbers)
- Working with tables (rows ⁵⁵55555555⁵⁵555555⁵⁵555555⁵⁵& columns)
- Matrix operations

Example :

```
For i in range(3):  
    for j in range(3):  
        print("*", end=" "  
    print()
```

EXAMPLE

```
For l in range(3):  
    for j in range(2):  
        print(l, j)
```

Output :

00	11
01	20
10	21

CONCLUSION

- Nested loops are powerful for handling complex tasks.
Useful in patterns, grids, and multi-dimensional data.
Important concept in Python programming.

Thank You